

Hankel-Norm and Information Limits on Delay Detection by Low-Order State-Space Models, with Application to Graph Random Walks

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Abstract—An order- S linear time-invariant (LTI) state-space model reads a sequence and must decide whether the current sample repeats the one k steps earlier. This delay-and-compare primitive underlies cycle detection by random-walk graph learners. We show that exact detection requires $S \geq k + 1$ and a nilpotent (FIR) state matrix, and is impossible at every S for invertible dynamics such as S4D and Mamba. Because the lag- k delay has k unit Hankel singular values, for $S < k$ no linear probe improves on the zero probe in Hankel norm, and any probe explains at most a fraction S/k of the delayed sample’s variance. For Gaussian inputs this bounds every nonlinear readout, with detection divergence at most $\frac{1}{2} \ln \frac{k}{k-S}$. Optimal fits follow the S/k law, and a trained shift register switches on exactly at $S = k + 1$. Trained diagonal SSMs stay below the Gaussian ceiling but need $4.0\times$ the minimal state to reach an AUROC of 0.95. Freezing their poles at the optimal fit raises the AUROC at $S = k$ from 0.70 to 0.87, so most of this overhead lies in learning the dynamics. On circular skip-link graphs, non-backtracking return horizons predict when exact-count features separate classes.

Index Terms—State-space models, Hankel operators, model order reduction, detection, random walks on graphs.

I. INTRODUCTION

DECIDING whether a signal repeats itself after a lag k is a basic detection task: it is echo detection, and it is the primitive that sequence models must implement to recall earlier inputs. Structured state-space models (SSMs) [1]–[3] implement it with a recursive filter of order S , and much of their design is about approximating delays with few states [4], [5]. That the pure delay z^{-k} has McMillan degree k is classical realisation theory [6]–[8]. What is missing is a *rate*: how much of the delayed sample can an order- S state carry when $S < k$, and what does that imply for detection by an arbitrary nonlinear readout? Related limits for recurrent models concern memory decay and optimisation [9], copying and associative recall [10], [11], and state tracking [12].

Our application is graph learning. Message-passing GNNs are bounded by the 1-WL test [13], [14] and cannot separate the regular circular skip-link (CSL) graphs [15], [16]. Walk-based learners read random walks as sequences [17]–[19], and random node features [20], [21] turn walk revisits, and hence closed walks, into token repetitions. SSMs are now used as the sequence backbone of such models [18], [22], [23].

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Code and data: <https://github.com/The-MathKing/ITE>.

Our contributions are: (i) *exact limits*: exact lag- k detection needs $S \geq k + 1$ with nilpotent dynamics and is impossible for invertible dynamics at any S ; (ii) *rates*: for $S < k$, no linear probe beats the zero probe in Hankel norm, the explained variance is at most S/k , and for Gaussian inputs this bounds the divergence available to every readout; (iii) *graph consequence*: exact non-backtracking (NB) return horizons $W^*(r)$ at which window features separate each CSL class; and (iv) *validation*: optimal fits, a shift-register control, frozen fitted poles and seeded experiments that separate the information limit from the optimisation overhead of trained SSMs.

II. PROBLEM SETTING

A sequence model with real state $s_t \in \mathbb{R}^S$ reads tokens $x_t \in \mathbb{R}^d$:

$$s_t = A_t s_{t-1} + B_t x_t, \quad \hat{y}_t = g(s_t, x_t), \quad (1)$$

with an arbitrary readout g . It is *LTI* if $A_t \equiv A$ and $B_t \equiv B$. The *lag- k repeat* label is $y_t = 1[x_t = x_{t-k}]$. In the graph setting, $v_0, v_1, \dots$ is an NB walk ($v_{t+1} \neq v_{t-1}$), each walk draws fresh features $f_v \sim \mathcal{N}(0, I_d)$ i.i.d. over nodes, and $x_t = f_{v_t}$. Then $y_t = 1[v_t = v_{t-k}]$ almost surely, i.e. $y_t = 1$ exactly when the last k steps form a closed NB walk. Their frequencies follow from powers of the Hashimoto (NB) operator [24], [25].

III. LIMITS

Proposition 1 (Exact detection). *Let (A, B) be LTI and g arbitrary. (a) If $S \leq k$, or (b) if A is invertible (any S), then there are two token sequences with the same (s_t, x_t) and $y_t = 1, y'_t = 0$. (c) If $S \geq k + 1$, the shift register $s_t = (w^\top x_t, \dots, w^\top x_{t-S+1})$, which has nilpotent A , detects y_t exactly for almost every w .*

Proof: Fix $\eta \neq 0$ and let $p(\lambda) = \sum_{i \leq q} p_i \lambda^i$ be the minimal polynomial of A on the Krylov space of $B\eta$, so $q \leq S$. Any polynomial $\pi(\lambda) = \sum_{j \geq 1} \pi_j \lambda^j$ with $\pi(A)B\eta = 0$ and $\pi_k \neq 0$ gives a confusable pair. Take a sequence with $x_{t-k} = x_t$ and set $x'_{t-j} = x_{t-j} + (\pi_j/\pi_k)\eta$ for $j \geq 1$. Then $s_t - s'_t = -\pi_k^{-1}\pi(A)B\eta = 0$ and $x'_{t-k} \neq x_t$. (a) If $q < k$, take $\pi = \lambda^{k-q}p$. If $q = k$ and $p = \lambda^k$, take $\pi = \lambda^k$. Otherwise let $i < k$ be the largest index with $p_i \neq 0$ and take $\pi = \lambda^{k-i}p$. (b) If A is invertible then $p_0 \neq 0$; take $\pi = \lambda^k p$. (c) The register holds $w^\top x_{t-k}$, and for continuous features $w^\top x_t = w^\top x_{t-k}$ iff $x_t = x_{t-k}$ almost surely. ■

Every S4D or Mamba discretisation $A = \exp(\Delta\Lambda)$ is invertible, so exactness is never available to them. The relevant

question is how well an order- S state can carry the lag- k sample. A linear probe $z_t = c^\top s_t + \beta^\top x_t$ estimates $w^\top x_{t-k}$ with $\|w\| = 1$. Its error $z_t - w^\top x_{t-k} = \sum_{m \geq 0} e_m x_{t-m}$ has taps $e_m = c^\top A^m B - \delta_{m,k} w^\top$ for $m \geq 1$ and transfer function $E(z) = \sum_{m \geq 1} e_m z^{-m}$, with $\|E(e^{j\omega})\|$ the Euclidean norm of the $1 \times d$ row. Let $\mathcal{H}_k(f)$ be the $k \times kd$ block-Hankel section with (i, j) block f_{i+j+1} , $0 \leq i, j < k$.

Theorem 1 (Hankel limits). *If $S < k$, every LTI probe satisfies*

$$\|\mathcal{H}_k(e)\|_2 \geq 1, \quad \text{hence} \quad \sup_\omega \|E(e^{j\omega})\| \geq 1, \quad (2)$$

$$\|\mathcal{H}_k(e)\|_F^2 = \sum_{m=1}^{2k-1} \min(m, 2k-m) \|e_m\|^2 \geq k - S. \quad (3)$$

Hence, for i.i.d. unit-variance tokens, $\mathbb{E}[(z_t - w^\top x_{t-k})^2] \geq 1 - S/k$: the probe explains at most a fraction S/k of the variance. For $S = k$ the infimum of the error is 0, but it is not attained.

Proof: Since $h_{i+j+1} = (c^\top A^i)(A^{j+1}B)$, we have $\text{rank } \mathcal{H}_k(h) \leq S$. The target $\delta_{m,k} w^\top$ gives a block anti-diagonal matrix with k unit singular values. Eckart–Young–Mirsky [26] in the spectral and Frobenius norms gives $\|\mathcal{H}_k(e)\|_2 \geq \sigma_{S+1} = 1$ and $\|\mathcal{H}_k(e)\|_F^2 \geq \sum_{i>S} \sigma_i^2 = k - S$. A finite section is bounded by the Hankel operator, which is a compression of the Laurent operator of E ; this gives (2), and equality in the limit is Nehari’s theorem [27]. The tap e_m occurs $\min(m, 2k - m) \leq k$ times, which gives (3) and the variance bound. For $S = k$, take $A = \gamma P$ with P the k -cycle permutation, $B = u_1 w^\top$, $c = \gamma^{-k} u_1$ and $\beta = -\gamma^{-k} w$. Then $e_m = \gamma^{m-k} w^\top$ for $m \in \{2k, 3k, \dots\}$ and 0 otherwise, so the error is $\gamma^{2k}/(1 - \gamma^{2k}) \rightarrow 0$ as $\gamma \rightarrow 0$. ■

Bound (2) says that for $S < k$ no LTI probe beats outputting zero in Hankel norm: all Hankel singular values of z^{-k} equal one, so its optimal Hankel-norm approximant of degree $< k$ is zero [7], [8]. Bound (3) is the rate. Under Gaussian inputs it extends to every readout.

Corollary 1 (Any readout). *Let x_t be i.i.d. $\mathcal{N}(0, I_d)$ and $S < k$. For every measurable g , the minimum mean-squared error of x_{t-k} given (s_t, x_t) is at least $d - S/k$. For H_1 : $x_t = x_{t-k}$ against H_0 : x_t fresh, the observations satisfy $\text{KL}(P_1 \| P_0) \leq \frac{1}{2} \ln \frac{k}{k-S}$, so no detector exceeds balanced accuracy $\frac{1}{2} + \frac{1}{2} \left(\frac{1}{4} \ln \frac{k}{k-S} \right)^{1/2}$.*

Proof: For jointly Gaussian variables the conditional mean is linear. Stacking d probes gives a target with kd unit Hankel singular values, and (3) yields total error $\geq d - S/k$. Next, (s_t, x_t) is a function of (s_{t-1}, x_t) . These are independent under H_0 and have the same marginals under H_1 , so $\text{KL} = I(s_{t-1}; x_{t-k}) = -\frac{1}{2} \sum_i \ln(1 - \lambda_i)$, where λ_i are the squared canonical correlations. Applying the bound to s_{t-1} (the same $k \times k$ Hankel count) gives $\sum_i \lambda_i \leq S/k$. By convexity the maximum puts all mass on one λ_i . Data processing and Pinsker’s inequality complete the proof. ■

Remark 1. The bounds limit *precision*, not information. For $S < k$ the state can still correlate with the lagged sample, so statistics pooled over many tokens remain informative. The near-optimal realisation at $S = k$ needs gains γ^{-k} and is exponentially ill-conditioned (cf. [9]). Input-dependent (A_t, B_t) break the factorisation used above, so we test them empirically.

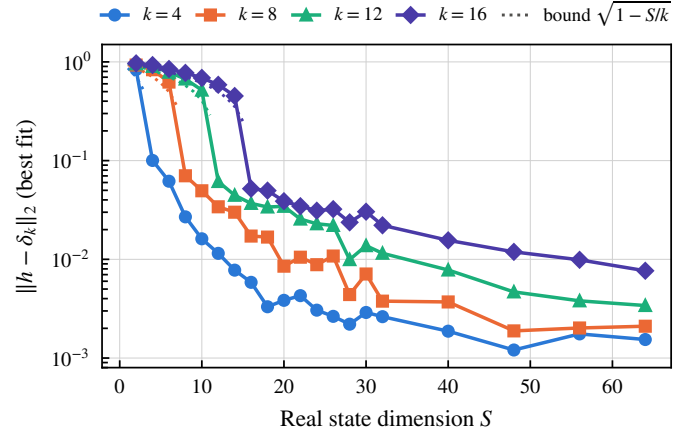


Fig. 1. Best gradient fit (four restarts, taps $m \leq 63$) of an S -state diagonal LTI impulse response to the lag- k delay. Dotted lines: the lower bound $\sqrt{1 - S/k}$ of Theorem 1 for $S < k$.

On walks, revisits make tokens dependent. Theorem 1(2) is distribution-free; the variance and divergence rates use the i.i.d. surrogate, which we check in Section IV.

NB return horizons. Let $p_G(\ell) = \Pr[v_\ell = v_0]$ for the stationary NB walk (uniform on directed edges). We compute it exactly from the Hashimoto operator [24]. For a family $\{G_r\}$, let $W^*(r)$ be the smallest W such that $\max_{\ell \leq W} |p_{G_r}(\ell) - p_{G_{r'}}(\ell)| > \tau$ for all $r' \neq r$. It is the shortest window of revisit frequencies that separates class r . For the ten CSL classes ($n = 41$, skips $r \in \{2, 3, 4, 5, 6, 9, 11, 12, 13, 16\}$), $W^*(r)$ ranges from 3 to 8 and is unchanged for $\tau \in [10^{-4}, 3 \cdot 10^{-3}]$ (Table I). W^* is a horizon for *exact* access. An order- S reader with $S < W^*$ still explains up to S/ℓ of the lag- ℓ variance, so pooled estimators can succeed below W^* at a sample cost that grows as S/W^* shrinks.

IV. EXPERIMENTS

Models. Our learned SSM is one diagonal complex layer with S4D-Lin initialisation [2]. It has $\lfloor S/2 \rfloor$ complex modes plus one real mode if S is odd, so its real state dimension is exactly S . The dynamics are $A = \exp(\Delta \Lambda)$ and $B_t = \Delta B$, with Δ learned per mode (LTI) or set to $\Delta_t = \text{softplus}(W_\Delta x_t + v_\Delta)$ (input-dependent step, the Mamba-style gate). The readout sees $[\Re s_t, \Im s_t, x_t]$ and comparison features $(U x_t - V s_t)^{\odot 2}$, followed by a two-layer MLP. We use $d = 8$ features, AdamW with a cosine schedule, and 3 seeds; intervals are 95% t -intervals. All runs are on CPU, and a script regenerates every number and figure in this letter.

Delay realisation. We fit an S -state diagonal impulse response to $\delta_{m,k}$ ($m \geq 1$), keeping the best of four restarts (Fig. 1). All 16 fits with $S < k$ respect $\|e\|_2 \geq \sqrt{1 - S/k}$, with a median excess MSE of 0.11 over the bound. The error first drops to about 0.1 at a median of $S = 1.0k$, so the rate of Theorem 1 is nearly attained by direct fitting.

Token-level detection. We train on NB walks ($T = 64$) over 256 random 4-regular graphs ($n = 24$, unions of two Hamiltonian cycles) and test on 64 unseen graphs. Fig. 2 shows that the test AUROC for y_t rises smoothly with S .

There is no sharp transition at $S = k$, where the AUROC is 0.69–0.72. The state needed for a given AUROC grows approximately linearly in k . Its median ratio to k is 2.0, 2.7, 4.0 and 4.0 at AUROC thresholds 0.8, 0.9, 0.95 and 0.99. Below the bound, all 22 LTI cells lie under the optimal AUROC of the Gaussian surrogate with explained variance S/k (Fig. 3). The surrogate is a fair model on these graphs. A Bayes-optimal table of the exact revisit indicators between x_t and lags $1, \dots, S-1$ predicts lag- k revisits at AUROC 0.49–0.52, so this short-lag walk structure carries no information about lag- k revisits. The input-dependent step changes the AUROC by -0.001 ± 0.002 (mean over cells; Wilcoxon $p = 0.005$). The change is consistent but negligible. Its learned Δ_t varies by only 1% across tokens, which is expected because random features give the gate nothing to select on.

Controls. Three controls separate the information limit from training effects (Fig. 3).

- *Shift register.* A trained shift register (nilpotent A , learned w , $k \in \{4, 8, 12, 16\}$) has AUROC 0.51–0.62 at $S = k$ and 1.00 at $S = k + 1$. This is the switch predicted by Proposition 1(c); the small excess over 0.5 at $S = k$ comes from walk structure among the stored lags.
- *Frozen poles.* We freeze the poles at the delay fit of Fig. 1 and train only B and the readout. The AUROC is 0.87 at $S = k$ and 0.97 at $S = 2k$ (mean over k), against 0.70 and 0.85 with learned poles. For $k \leq 12$ it is already 0.92–0.97 at $S = k$. At $k = 16$ it drops to 0.64 ± 0.26 , where the fitted poles are ill-conditioned (Remark 1).
- *Comparison features.* Removing them changes the $k = 8$ AUROC by at most 0.02 at any S .

The information limit is therefore nearly attainable at $S \approx k$. Most of the overhead of trained SSMs comes from learning the dynamics, not from what an order- S state can represent or from the readout.

CSL classification. CSL graphs are 4-regular and featureless, so 1-WL is provably at chance (10%). Each training example is one NB walk ($T = 128$) on a CSL graph, and the class is predicted from mean-pooled features. At test time, each graph is judged from 16 walks ($T = 256$). As an *exact-count reference* we use logistic regression on the exact revisit frequencies at lags $1, \dots, S$. It is not an upper bound. At $S = 2$ it is at chance, because NB walks cannot return at lags 1–2, while the LTI model reaches 24.0%. The reference reaches 98.3%. For all 10 classes, its onset of 80% class accuracy is the first tested $S \geq W^*(r)$ (Table I). The trained models reach $60.4 \pm 7.9\%$ (LTI) and $55.2 \pm 8.5\%$ (input-dependent step) at $S = 64$. The largest tested S gives the best accuracy for both, and the curves are still rising. Neither remedy we tried closes the gap:

- A per-lag revisit loss gives $56.2 \pm 13.7\%$ at $S = 64$.
- Evaluating at the training length ($T = 128$) instead of $T = 256$ lowers accuracy by 1.9 points.

The input-dependent step is -1.7 ± 1.1 points from LTI (paired over S and seeds). By the token-level controls, the likely bottleneck is learning precise poles for several class-specific lags from a pooled graph label, but we do not isolate it here. Class $r = 2$ ($W^* = 3$) is learned at $S = 2$ only by pooling:

TABLE I
CSL HORIZONS $W^*(r)$ AND PER-CLASS ONSETS: THE SMALLEST TESTED S WITH CLASS ACCURACY $\geq 80\%$ (MEAN OF 3 SEEDS; —: NOT REACHED FOR $S \leq 64$).

r	2	3	4	5	6	9	11	12	13	16
$W^*(r)$	3	4	5	6	7	8	7	8	5	7
Exact counts	4	4	6	6	8	8	8	8	6	8
LTI	2	6	—	—	—	—	—	—	32	—
Input-dep. Δ_t	2	8	—	—	—	—	—	—	64	—

with one walk its accuracy is 53.7%, and with 16 walks it is 86.3%.

V. CONCLUSION

The pure delay has k unit Hankel singular values. For an order- S LTI state this gives three limits: exact detection needs $S \geq k + 1$ and nilpotent dynamics, the explained variance of the lagged sample is at most S/k , and for Gaussian inputs every readout has divergence at most $\frac{1}{2} \ln \frac{k}{k-S}$. Optimal fits and a shift register attain these limits. Trained diagonal SSMs respect them, but they need several times the minimal state, and an input-dependent step does not change this. Lower bounds for input-dependent dynamics and better-conditioned delay parameterisations are natural next steps.

REFERENCES

- [1] A. Gu, K. Goel, and C. Ré, “Efficiently modeling long sequences with structured state spaces,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.
- [2] A. Gu, A. Gupta, K. Goel, and C. Ré, “On the parameterization and initialization of diagonal state space models,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022.
- [3] A. Gu and T. Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” in *Proc. Conf. Lang. Model. (COLM)*, 2024.
- [4] A. R. Voelker, I. Kajić, and C. Eliasmith, “Legendre memory units: Continuous-time representation in recurrent neural networks,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2019.
- [5] A. Gu, T. Dao, S. Ermon, A. Rudra, and C. Ré, “HiPPO: Recurrent memory with optimal polynomial projections,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2020.
- [6] B. L. Ho and R. E. Kalman, “Effective construction of linear state-variable models from input/output functions,” *Regelungstechnik*, vol. 14, no. 12, pp. 545–548, 1966.
- [7] K. Glover, “All optimal Hankel-norm approximations of linear multi-variable systems and their L^∞ -error bounds,” *Int. J. Control*, vol. 39, no. 6, pp. 1115–1193, 1984.
- [8] V. M. Adamjan, D. Z. Arov, and M. G. Kreĭn, “Analytic properties of Schmidt pairs for a Hankel operator and the generalized Schur–Takagi problem,” *Math. USSR-Sbornik*, vol. 15, no. 1, pp. 31–73, 1971.
- [9] Z. Li, J. Han, W. E, and Q. Li, “On the curse of memory in recurrent neural networks: Approximation and optimization analysis,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021.
- [10] S. Jelassi, D. Brandfonbrener, S. M. Kakade, and E. Malach, “Repeat after me: Transformers are better than state space models at copying,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2024.
- [11] S. Arora, S. Eyuboglu, A. Timalina, I. Johnson, M. Poli, J. Zou, A. Rudra, and C. Ré, “Zoology: Measuring and improving recall in efficient language models,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.
- [12] W. Merrill, J. Petty, and A. Sabharwal, “The illusion of state in state-space models,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2024.
- [13] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, “How powerful are graph neural networks?” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019.
- [14] C. Morris, M. Ritzert, M. Fey, W. L. Hamilton, J. E. Lenssen, G. Rattan, and M. Grohe, “Weisfeiler and Leman go neural: Higher-order graph neural networks,” in *Proc. AAAI Conf. Artif. Intell.*, 2019.

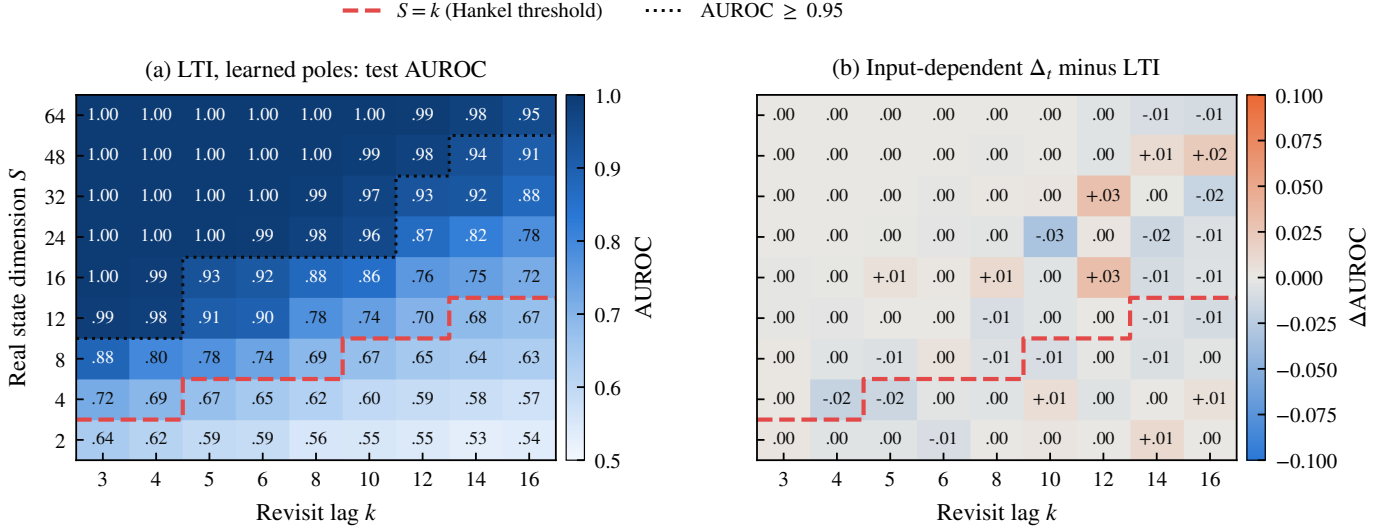


Fig. 2. Lag- k revisit detection on NB walks over random 4-regular graphs ($n = 24$, $T = 64$, $d = 8$), mean of 3 seeds. (a) Test AUROC of the LTI model; the dotted line marks the first S with $\text{AUROC} \geq 0.95$. (b) Input-dependent step minus LTI. Red dashed line: $S = k$.

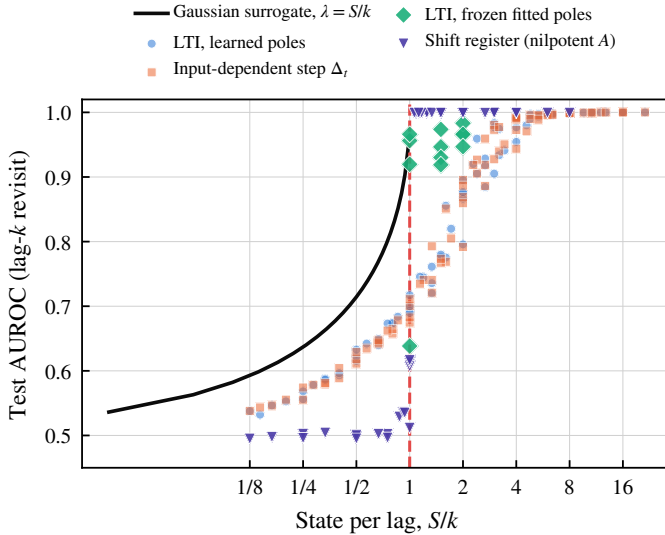


Fig. 3. Test AUROC against S/k (all k , seed means). Solid curve: optimal AUROC of the i.i.d. Gaussian surrogate with explained variance S/k (Corollary 1). Red dashed line: $S = k$.

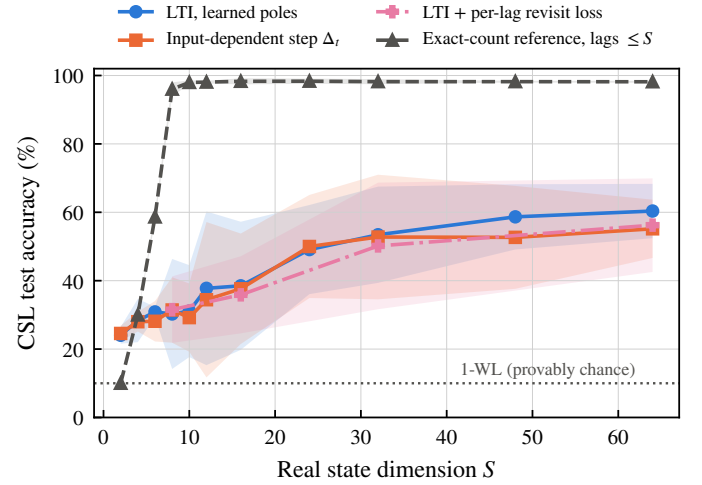


Fig. 4. CSL test accuracy against S (mean and 95% interval over 3 seeds; test graphs are the ten CSL graphs with fresh walks and features). The exact-count reference uses revisit frequencies at lags $\leq S$.

- graph neural networks,” in *Proc. SIAM Int. Conf. Data Mining (SDM)*, 2021.
- [15] R. L. Murphy, B. Srinivasan, V. Rao, and B. Ribeiro, “Relational pooling for graph representations,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2019.
- [16] V. P. Dwivedi, C. K. Joshi, A. T. Luu, T. Laurent, Y. Bengio, and X. Bresson, “Benchmarking graph neural networks,” *J. Mach. Learn. Res.*, vol. 24, no. 43, pp. 1–48, 2023.
- [17] J. Tönshoff, M. Ritzert, H. Wolf, and M. Grohe, “Walking out of the Weisfeiler Leman hierarchy: Graph learning beyond message passing,” *Trans. Mach. Learn. Res.*, 2023.
- [18] D. Chen, T. H. Schulz, and K. Borgwardt, “Learning long range dependencies on graphs via random walks,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025.
- [19] J. Kim, O. Zaghen, A. Suleymanzade, Y. Ryou, and S. Hong, “Revisiting random walks for learning on graphs,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025.
- [20] R. Abboud, I. I. Ceylan, M. Grohe, and T. Lukasiewicz, “The surprising power of graph neural networks with random node initialization,” in *Proc. Int. Joint Conf. Artif. Intell. (IJCAI)*, 2021.
- [21] R. Sato, M. Yamada, and H. Kashima, “Random features strengthen
- [22] C. Wang, O. Tsepa, J. Ma, and B. Wang, “Graph-Mamba: Towards long-range graph sequence modeling with selective state spaces,” *arXiv preprint arXiv:2402.00789*, 2024.
- [23] A. Behrouz and F. Hashemi, “Graph Mamba: Towards learning on graphs with state space models,” in *Proc. ACM SIGKDD Conf. Knowl. Discov. Data Mining (KDD)*, 2024.
- [24] K.-i. Hashimoto, “Zeta functions of finite graphs and representations of p -adic groups,” in *Automorphic Forms and Geometry of Arithmetic Varieties*, ser. Adv. Stud. Pure Math. Academic Press, 1989, vol. 15, pp. 211–280.
- [25] N. Alon, I. Benjamini, E. Lubetzky, and S. Sodin, “Non-backtracking random walks mix faster,” *Commun. Contemp. Math.*, vol. 9, no. 4, pp. 585–603, 2007.
- [26] C. Eckart and G. Young, “The approximation of one matrix by another of lower rank,” *Psychometrika*, vol. 1, no. 3, pp. 211–218, 1936.
- [27] Z. Nehari, “On bounded bilinear forms,” *Ann. Math.*, vol. 65, no. 1, pp. 153–162, 1957.